Data

The data used cover three domains: income, home values, and retail environments. Income and home value data is used to characterize a neighborhood's gentrification status, following a method developed by Bunten et al. (2023). Retail environment data is then used to characterize changes in the establishments present in a ZIP code over time. The consistent unit of measurement across all datasets is a ZIP code-year.

The baseline population for this study consists of all ZIP codes in each of six core-based statistical areas (CBSAs): Chicago, New York City, Boston, Los Angeles, Washington D.C., and San Francisco. These areas were chosen based on a combination of population size and reports of extensive gentrification in these areas. While all ZIP codes in each CBSA are used when computing the CBSA's distribution of home values and income, only ZIP codes in central counties are considered eligible to undergo gentrification in this study.

Income

Income data come from the Statistics of Income files provided by the IRS. These files list ZIP code level totals for the number of returns filed in each of six income brackets, as well as the total dollar amount of income falling into each bracket. These files are provided in a standardized format from 2011 to 2022. Earlier years use varying delineations of the income brackets, and for this reason, the years 2011 to 2022 were chosen for this study.

ZIP codes are matched to each CBSA using the HUD CBSA-ZIP crosswalk, accessed through the HUD User API. Median incomes for each ZIP code are computed using interpolation assuming a uniform distribution within each income bin. CBSA-level distributions are computed using R's "splinebins" package, which specifically handles calculations for binned datasets. The CBSA-level distribution for each year is then used to find each ZIP code's income percentile ranking within its CBSA.

Home value

Home value data is sourced from the Zillow Home Value Index (ZHVI), which provides an estimate of the value of single-family homes in the 35th to 65th percentile of each ZIP code tablulation area (ZCTA). ZCTAs differ slightly from ZIP codes, so a crosswalk was used to match ZIP codes to their associated ZCTA in order for the income and retail data to be merged with the home value data. The ZHVI is provided on a monthly basis, and annual averages per ZIP code were computed for the purpose of this study. Similarly to the income data, the home value data for each ZIP code-year was assigned a percentile ranking based on the distribution of the CBSA that the ZIP code belongs to in that year.

Retail distribution

The retail environment of each ZIP code was assessed using data from the National Neighborhood Data Archive, which collects and publishes data on counts of businesses in different categories based on NAICS codes. The categories of interest in this study thus far are: dollar stores, coffee shops and cafés, delis, and the total number of arts, entertainment and leisure establishments. In the future, more categories will be added to give a more comprehensive picture of a ZIP code's retail environment.

Figures 2 through 4 show the shares of different types of retail establishments in high-income, moderate-income, and low-income ZIP codes in 2011. Here, high-income ZIP codes are defined as those in the top twenty-fifth percentile, low-income as those within the bottom fortieth percentile, and moderate-income as those in between.

From these figures, some clear preliminary differences can be seen in the retail distributions of low, moderate, and high-income areas. High-income areas have more coffee shops and entertainment-related establishments, while low-income areas have more dollar stores and fast-food restaurants.

Identification of gentrification

From among the ZIP codes in each CBSA, those designated as gentrifiable in any given year are those which satisfy the following two conditions: (1) the median income of the ZIP code is below the fortieth percentile relative to the CBSA, and (2) the ZIP code lies in one of the central counties of the CBSA. Those ZIP codes identified as gentrifying in any given year are those which were in a previous year identified as gentrifiable, and exhibit a twenty-five percentile point gap between their average home value and their median income.

Figure 1 shows the share of ZIP codes in each CBSA in 2011 that are considered eligible for gentrification according to the two conditions outlined above.

Figure 1: Share of Gentrifiable ZIP Codes in 2011

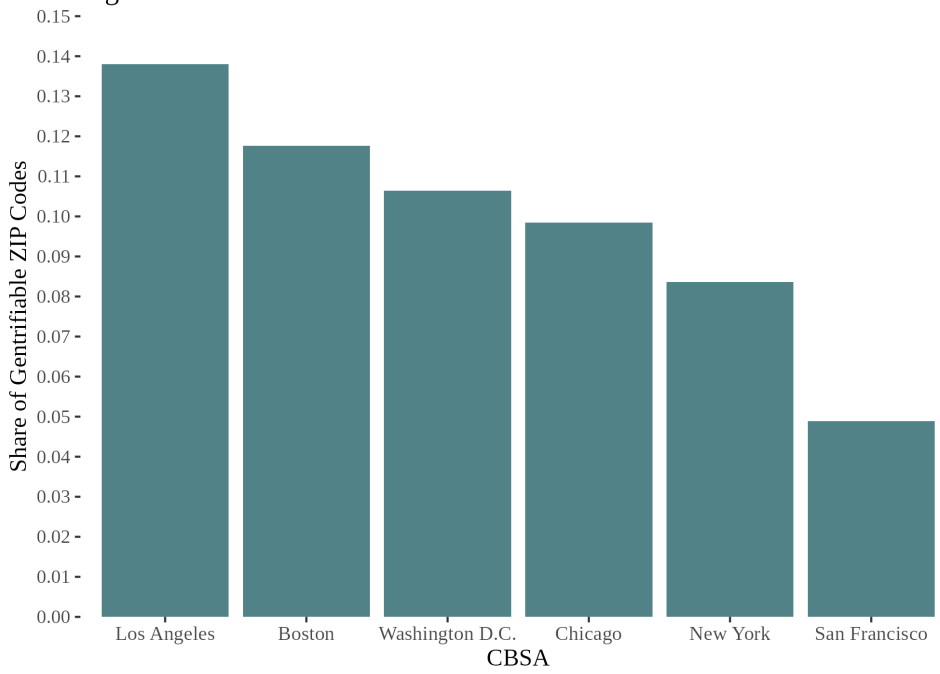
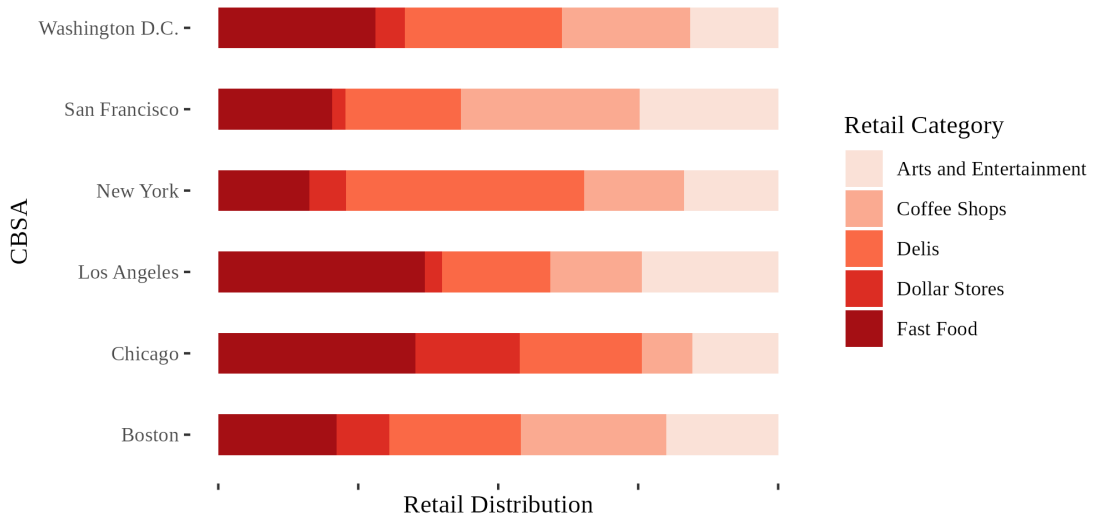
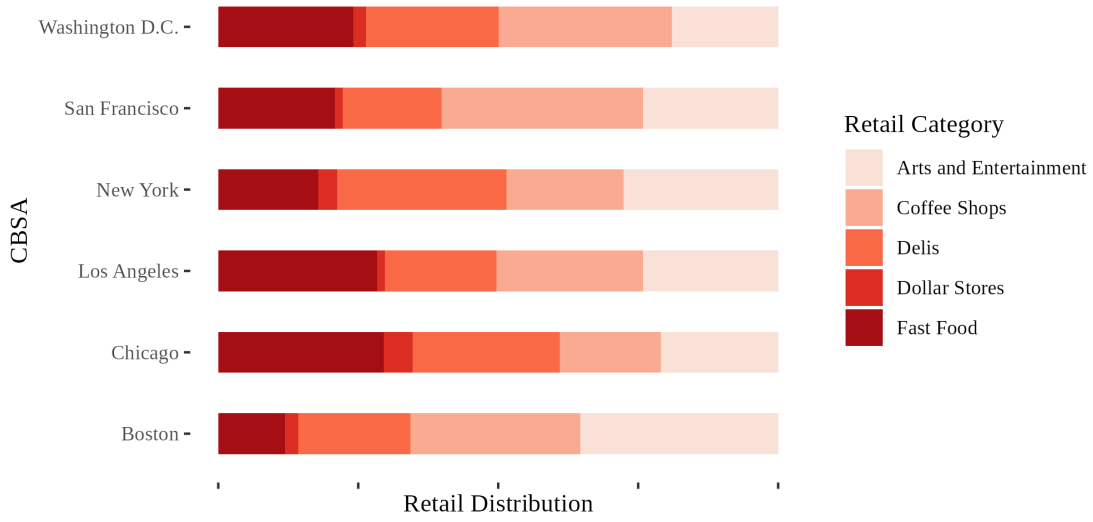


Figure 2: Low-Income Retail Distribution by CBSA in 2011



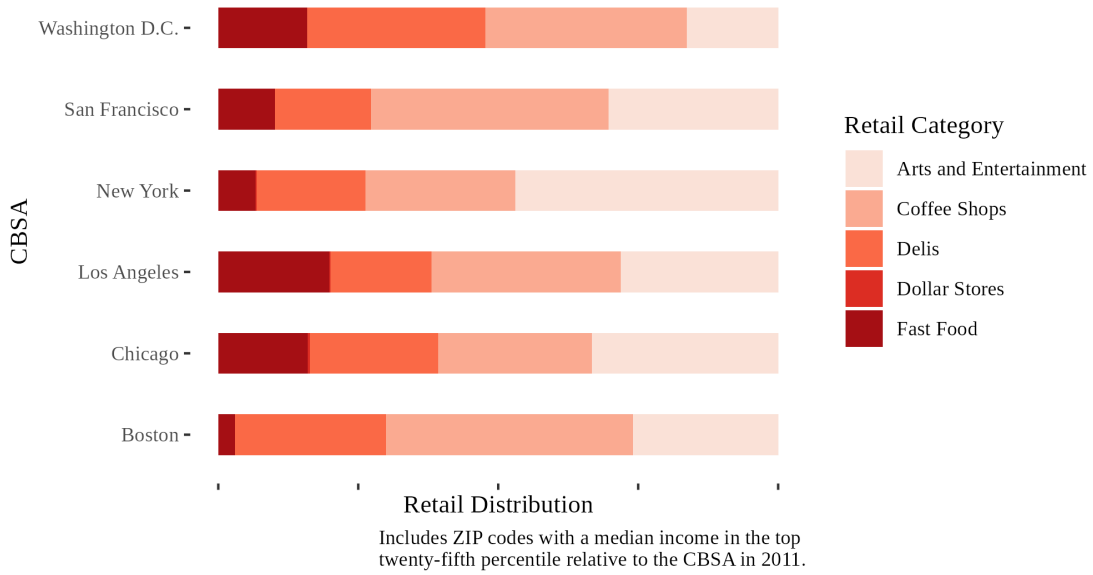
Includes ZIP codes with a median income in the bottom fortieth percentile relative to the CBSA in 2011.

Figure 3: Moderate-Income Retail Distribution by CBSA in 2011



Includes ZIP codes with a median income below the top twenty-fifth percentile and above the fortieth percentile relative to the CBSA in 2011.

Figure 4: High-Income Retail Distribution by CBSA in 2011



Includes ZIP codes with a median income in the top twenty-fifth percentile relative to the CBSA in 2011.

References

Housing Data. Zillow. (n.d.). https://www.zillow.com/research/data/

SOI tax stats - individual income tax statistics - ZIP code data (SOI). Internal Revenue Service. (n.d.). https://www.irs.gov/statistics/soi-tax-stats-individual-income-tax-statistics-zip-code-data-soi

University of Michigan. (n.d.). Data. NaNDA - National Neighborhood Data Archive. https://nanda.isr.umich.edu/data/